PHYS121 Integrated Science-Physics

W5 T2 Fluids (part 2)

References:

[1] David Halliday, Jearl Walker, Resnick Jearl, ‘Fundamentals of Physics’, (Wiley, 2018)
[2] Doug Giancoli, ‘Physics for Scientists and Engineers with modern physics’, (Pearson, 2009)
[3] Hugh D. Young, Roger A. Freedman, ‘University Physics with Modern Physics’, (Pearson, 2012)
And others specified when needed.

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Learning Outcomes

- Perform calculations of basic properties of fluid, i.e., about density, pressure, buoyant force.
- Apply the equation of continuity of ideal fluids.
- Analyze typical examples of application of Bernoulli's principle/equation.
- Explain the phenomenon of viscosity and perform simple calculation on it.

Bernoulli's Equation

- Figure 14-19 represents a tube through which an ideal fluid flows
- Applying the conservation of energy to the equal volumes of input and output fluid:

$$p_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = p_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2.$$

- The $\frac{1}{2}\rho v^2$ term is called the fluid's **kinetic energy density**

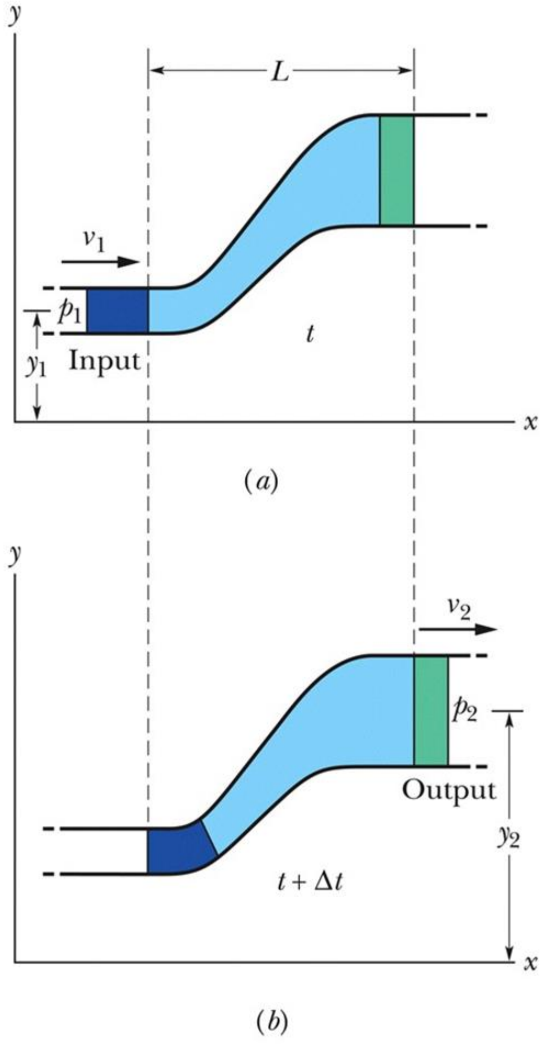


Figure 14-19

- Equivalent to Eq. 14-28, we can write:

$$p + \frac{1}{2}\rho v^2 + \rho g y = \text{a constant} \qquad \textbf{Equation (14-29)}$$

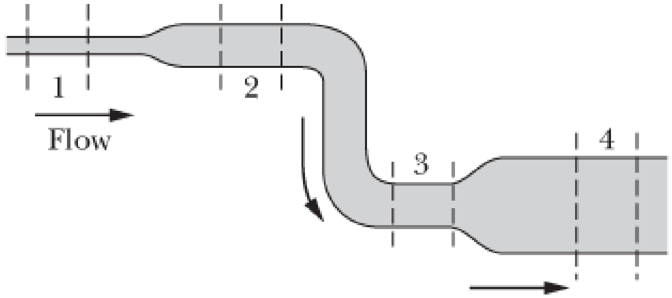
- These are both forms of **Bernoulli's Equation**
- Applying this for a fluid at rest we find Eq. 14-7
- Applying this for flow through a horizontal pipe:

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2, \qquad \textbf{Equation (14-30)}$$

If the speed of a fluid element increases as the element travels along a horizontal streamline, the pressure of the fluid must decrease, and conversely.

Checkpoint 4

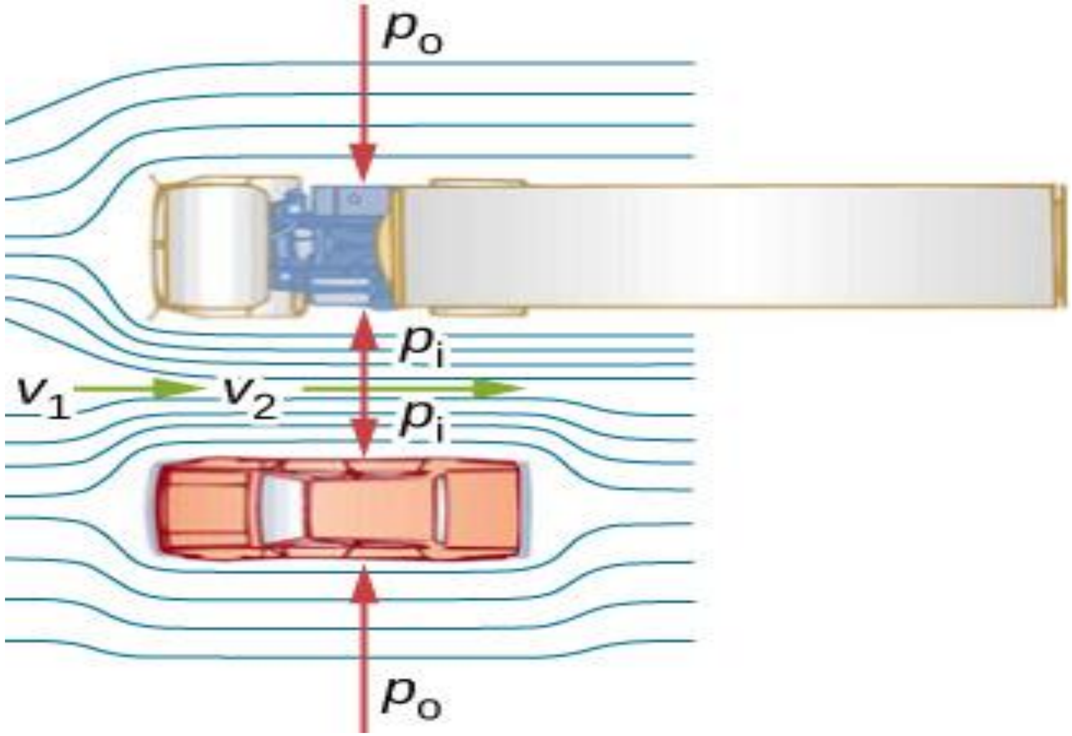
Water flows smoothly through the pipe shown in the figure, descending in the process. Rank the four numbered sections of pipe according to (a) the volume flow rate R_V through them, (b) the flow speed v through them, and (c) the water pressure p within them, greatest first.



Answer:

(a) all the same volume flow rate.
(b) 1, 2 & 3, 4.
(c) 4, 3, 2, 1.

FIGURE 14.29



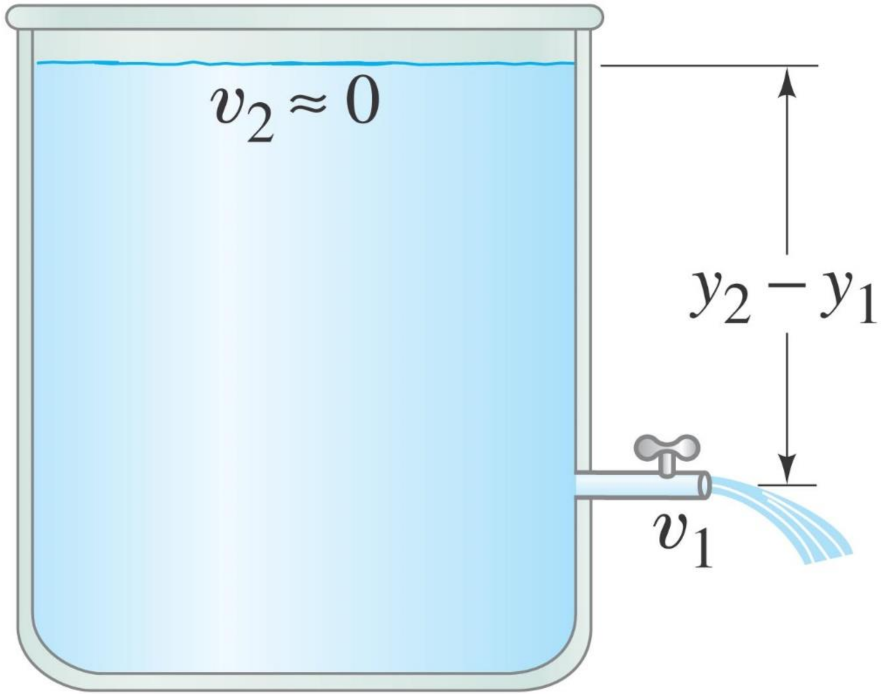
An overhead view of a car passing a truck on a highway. Air passing between the vehicles flows in a narrower channel and must increase its speed (v_2 is greater than v_1) [Why?], causing the pressure between them to drop (p_i is less than p_o). Greater pressure on the outside pushes the car and truck together.

Example: Flow and pressure in a hot-water heating system.

Water circulates throughout a house in a hot-water heating system. If the water is pumped at a speed of 0.5 m/s through a 4.0-cm-diameter pipe in the basement under a pressure of 3.0 atm (gauge pressure), what will be the flow speed and pressure in a 2.6-cm-diameter pipe on the second floor 5.0 m above? Assume the pipes do not divide into branches.

Applications of Bernoulli's Principle: Torricelli, Airplanes, Baseballs, TIA

Using Bernoulli's principle, we find that the speed of fluid coming from a **spigot** on an **open** tank is:



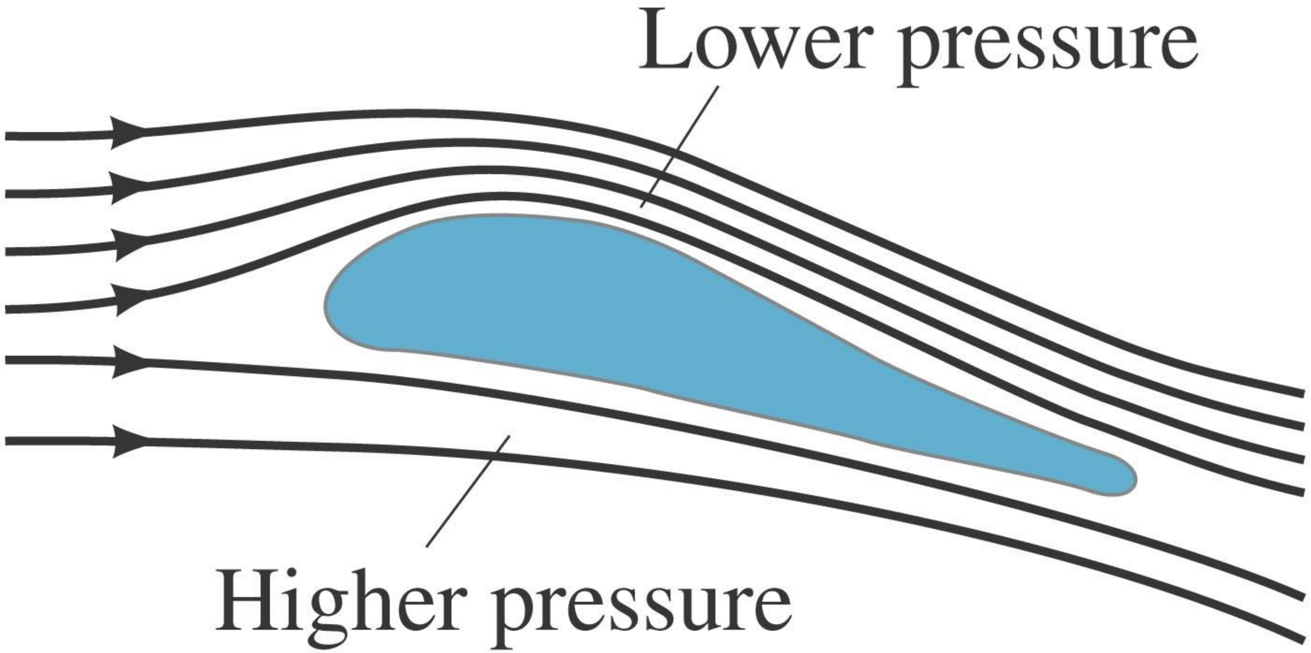
$$\tfrac{1}{2}\rho v_1^2 + \rho g y_1 = \rho g y_2$$

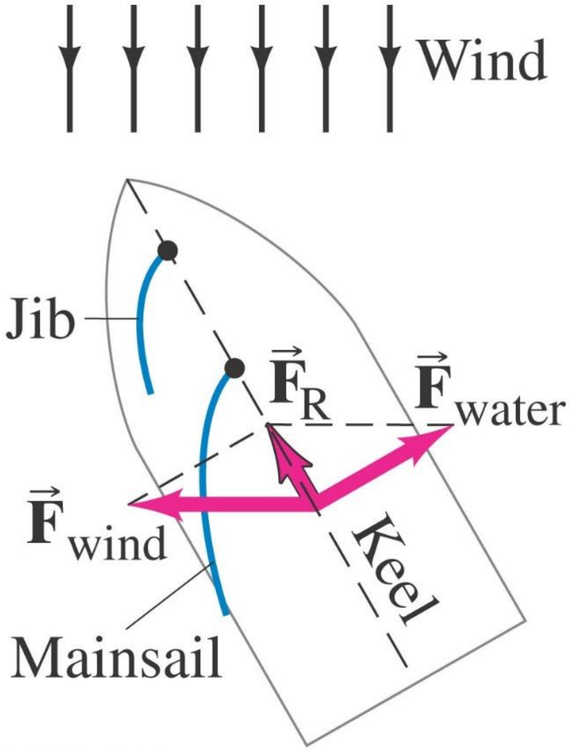
or

$$v_1 = \sqrt{2g(y_2 - y_1)}.$$

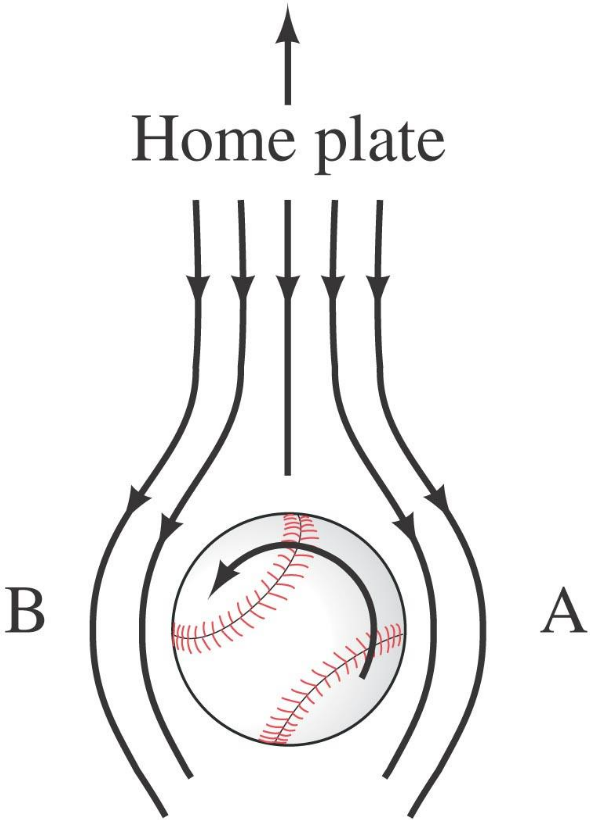
This is called **Torricelli's** theorem.

Lift on an airplane **wing** is due to the different air **speeds** and **pressures** on the two surfaces of the wing.

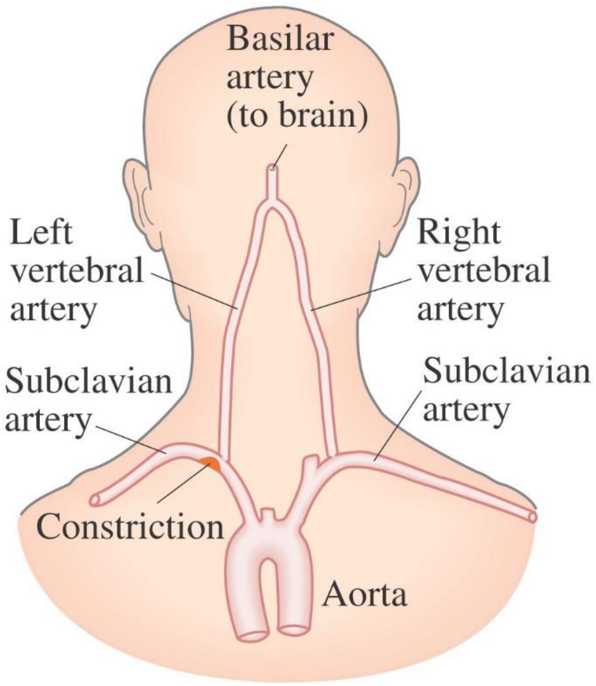




A sailboat can move against the wind, using the pressure differences on each side of the sail, and using the keel to keep from going sideways.

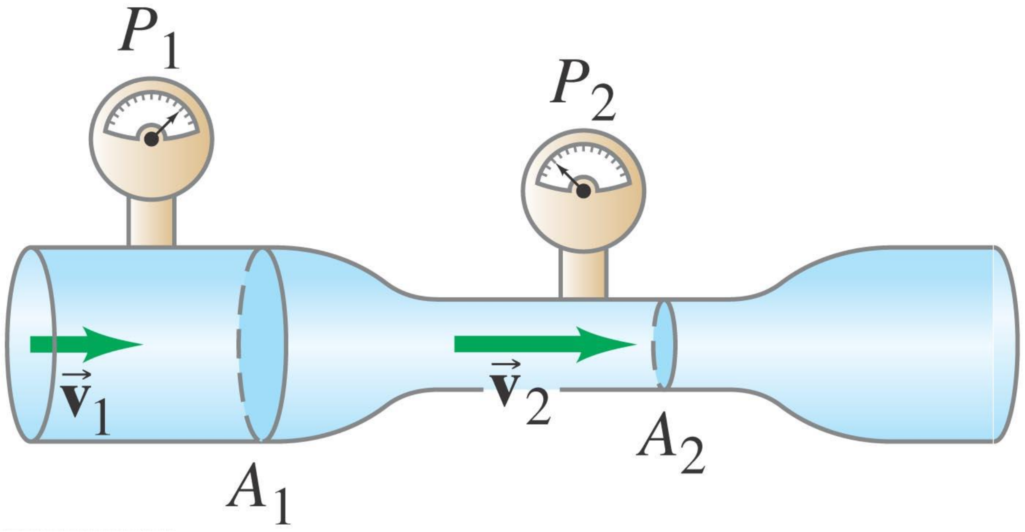


A ball's path will curve due to its spin, which results in the air speeds on the two sides of the ball not being equal; therefore there is a pressure difference.



A person with constricted arteries may experience a temporary lack of blood to the brain (TIA) as blood speeds up to get past the constriction, thereby reducing the pressure.

A venturi meter can be used to measure fluid flow by measuring pressure differences.



Viscosity

Real fluids have some **internal friction**, called **viscosity**.

The **viscosity** can be measured; it is found from the relation

$$F = \eta A \frac{v}{\ell}.$$

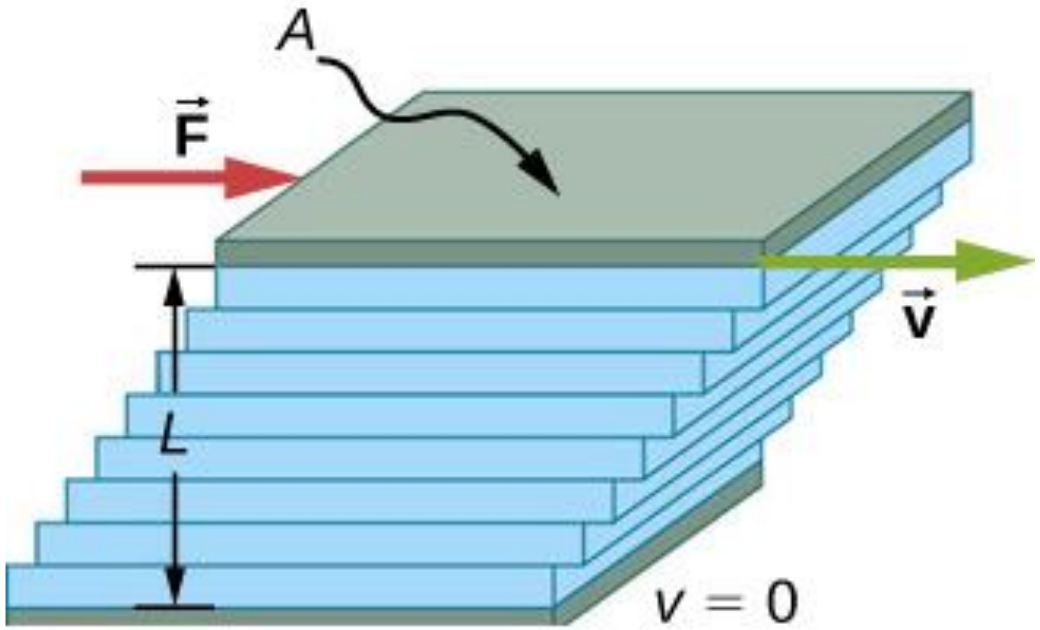


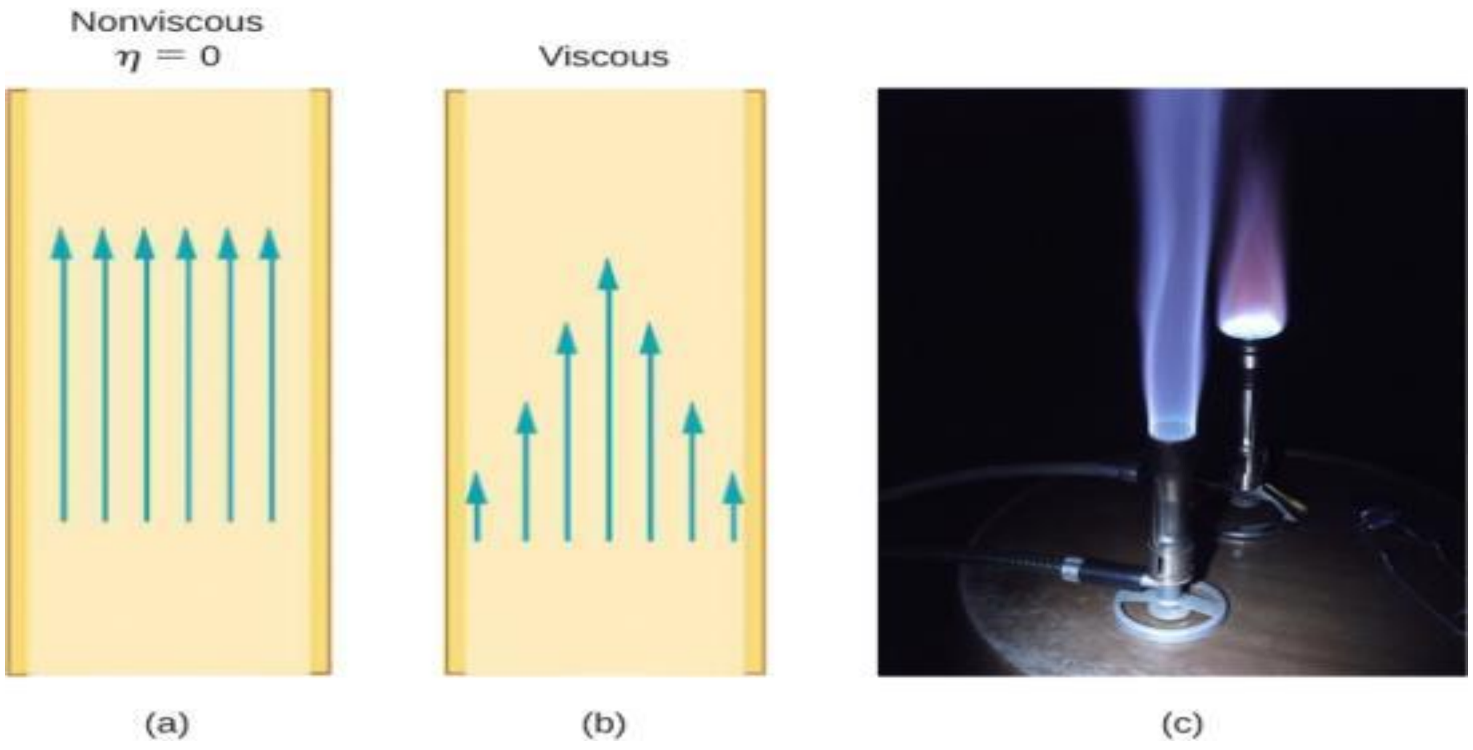
Fig. 14.36, from OpenStax textbook

TABLE 13–3
Coefficients of Viscosity

Fluid (temperature in °C)	Coefficient of Viscosity, η (Pa · s)†
Water (0°)	1.8×10^{-3}
(20°)	1.0×10^{-3}
(100°)	0.3×10^{-3}
Whole blood (37°)	$\approx 4 \times 10^{-3}$
Blood plasma (37°)	$\approx 1.5 \times 10^{-3}$
Ethyl alcohol (20°)	1.2×10^{-3}
Engine oil (30°) (SAE 10)	200×10^{-3}
Glycerine (20°)	1500×10^{-3}
Air (20°)	0.018×10^{-3}
Hydrogen (0°)	0.009×10^{-3}
Water vapor (100°)	0.013×10^{-3}

† 1 Pa · s = 10 P = 1000 cP.

FIGURE 14.37

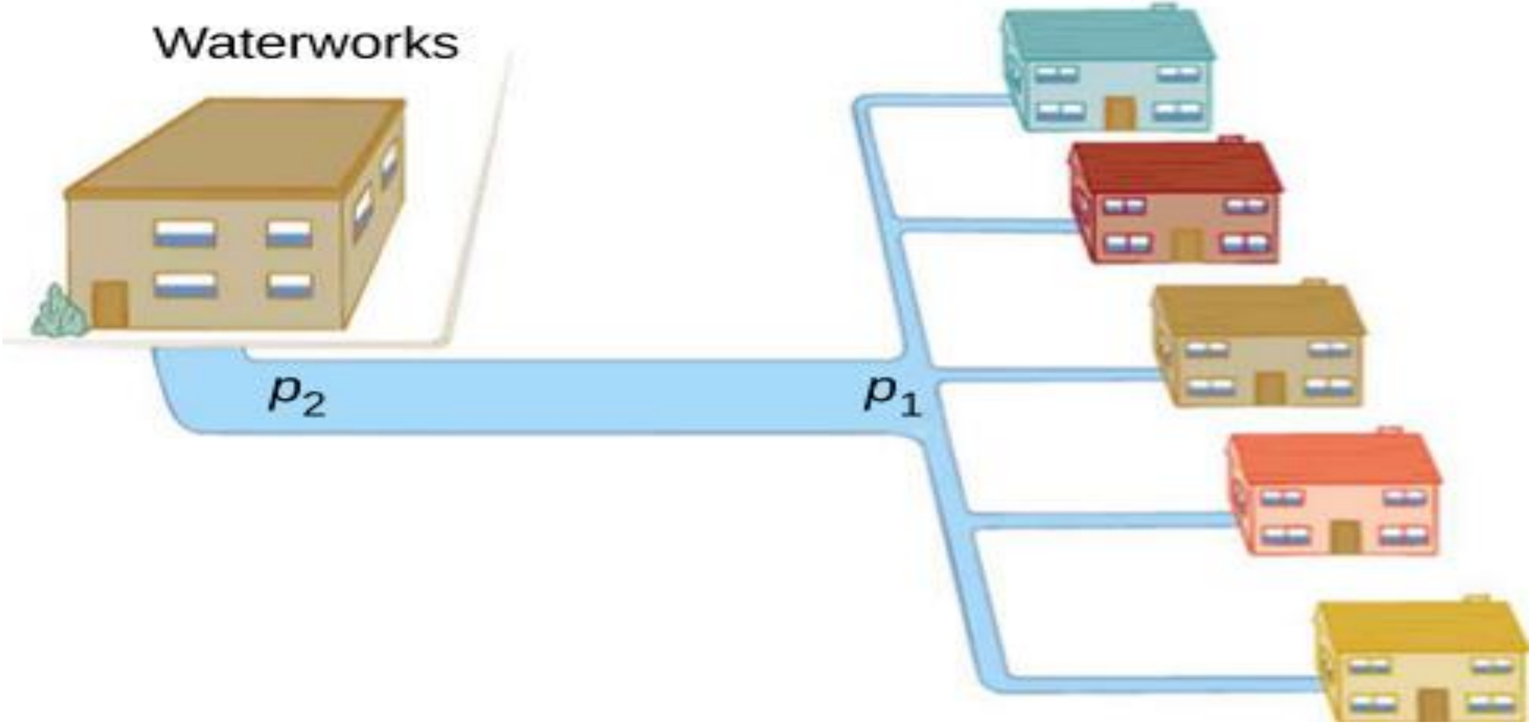


(a) If fluid flow in a tube has negligible resistance, the speed is the same all across the tube.

(b) When a viscous fluid flows through a tube, its speed at the walls is zero, increasing steadily to its maximum at the center of the tube.

(c) The shape of a Bunsen burner flame is due to the velocity profile across the tube. (credit c: modification of work by Jason Woodhead)

FIGURE 14.39



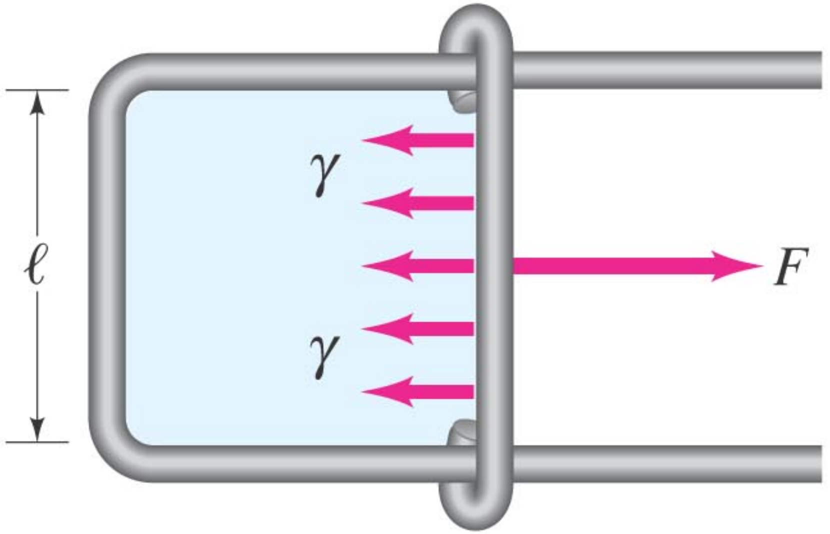
During times of heavy use, there is a significant pressure drop in a water main, and p_1 supplied to users is significantly less than p_2 created at the water works. If the flow is very small, then the pressure drop is negligible, and $p_2 \approx p_1$.

Surface Tension and Capillarity

The **surface** of a liquid at rest is not perfectly flat; it **curves** either up or down at the **walls** of the container. This is the result of **surface tension**, which makes the surface behave somewhat elastically.

The surface tension is defined as the force per unit length that acts perpendicular to the surface:

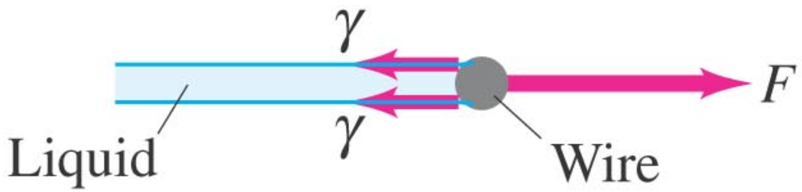
$$\gamma = \frac{F}{\ell}.$$



Top view

TABLE 13–4
Surface Tension of Some Substances

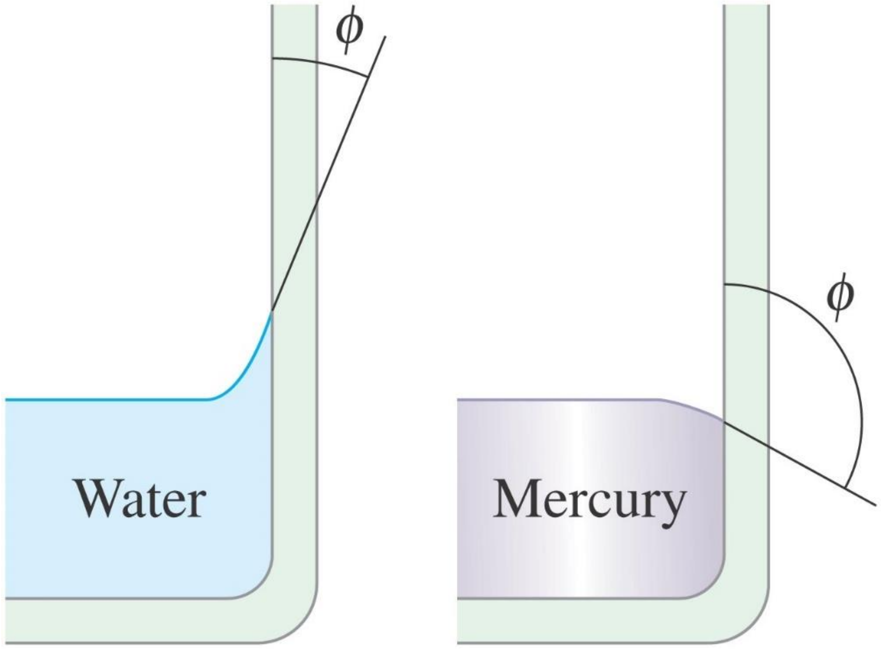
Substance (temperature in °C)	Surface Tension (N/m)
Mercury (20°)	0.44
Blood, whole (37°)	0.058
Blood, plasma (37°)	0.073
Alcohol, ethyl (20°)	0.023
Water (0°)	0.076
(20°)	0.072
(100°)	0.059
Benzene (20°)	0.029
Soap solution (20°)	≈0.025
Oxygen (−193°)	0.016



Edge view (magnified)

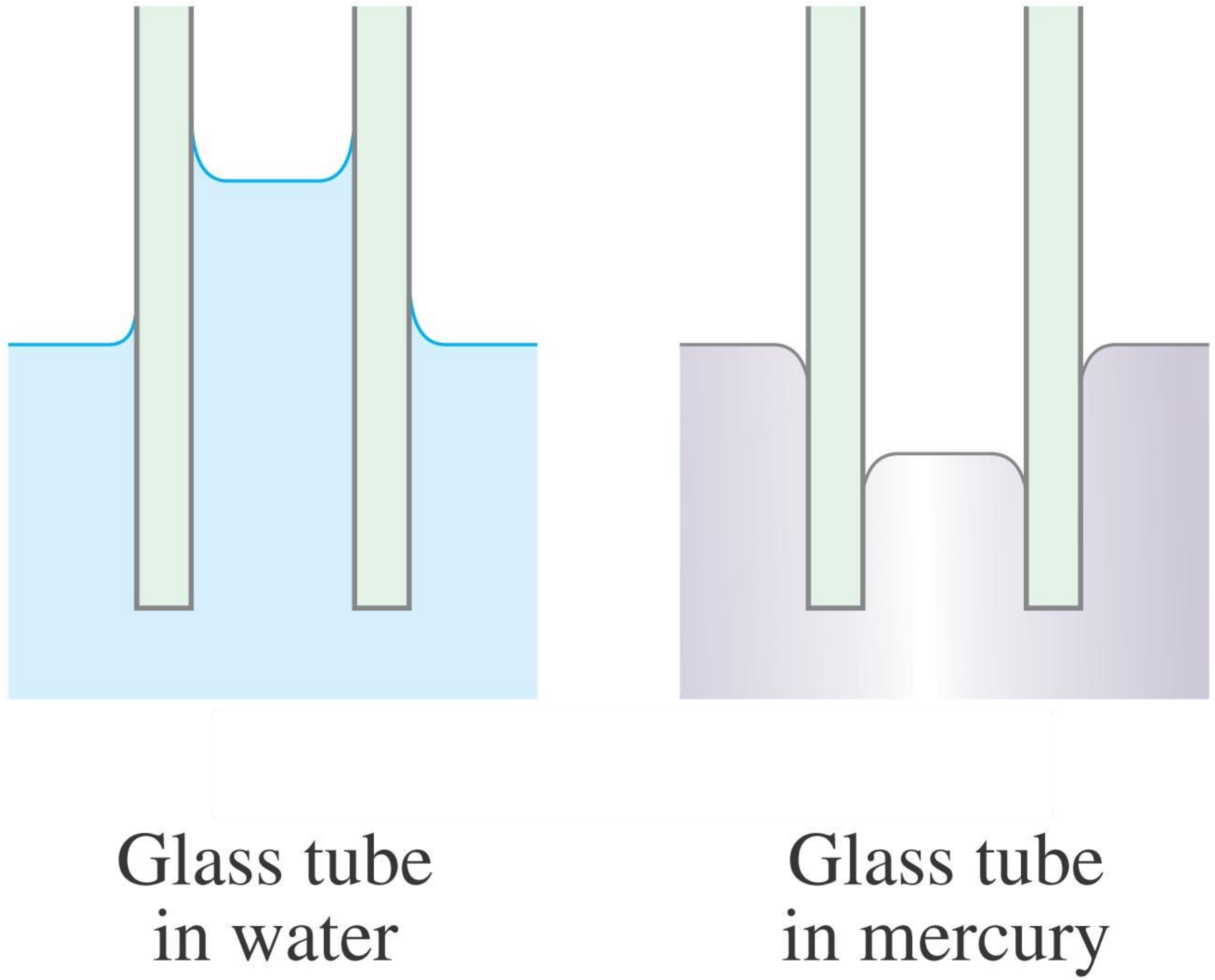
Because of surface tension, some objects more dense than water may not sink.

Soap and detergents lower the surface tension of water. This allows the water to penetrate materials more easily.



Water molecules are more strongly attracted to glass than they are to each other; just the opposite is true for mercury.

If a narrow tube is placed in a fluid, the fluid will exhibit capillarity.



Superfluid

https://www.youtube.com/watch?v=2Z6UJbwxBZI

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- Analyze typical examples of application of Bernoulli's principle/equation.
- Explain the phenomenon of viscosity and perform simple calculation on it.

$$p = p_0 + \rho g h \qquad F_b = m_f g$$

$$p_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = p_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2.$$

$$F = \eta A \frac{v}{\ell}.$$

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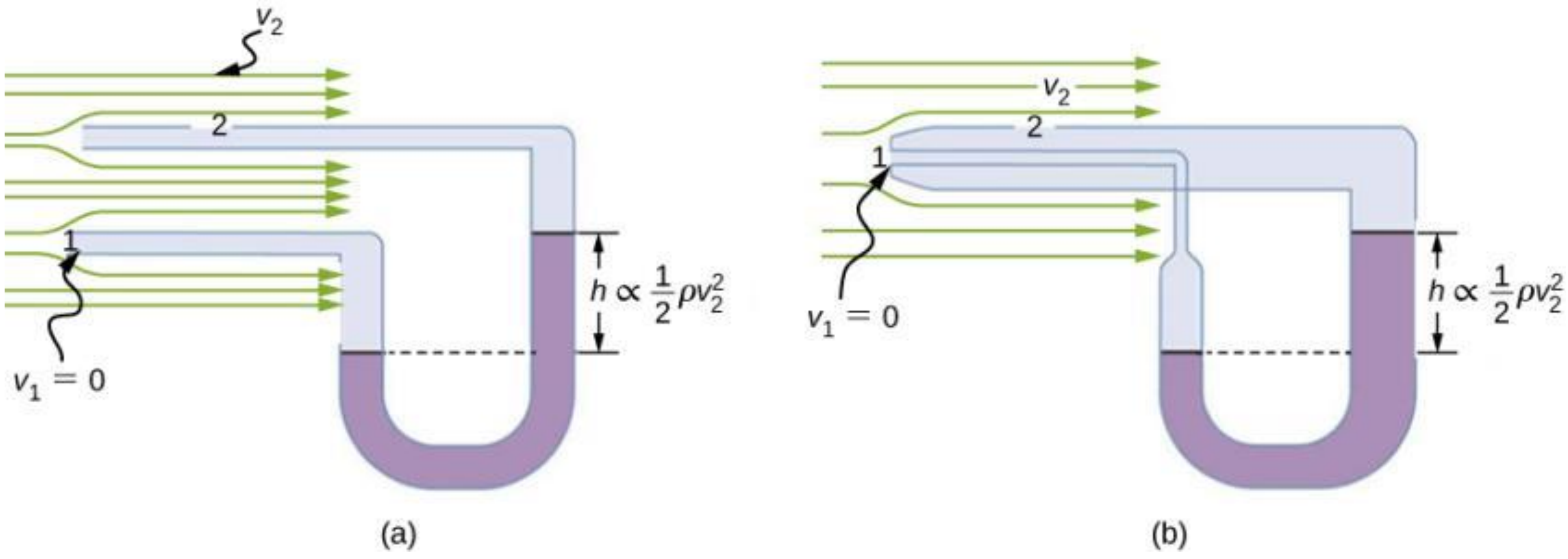
FIGURE 14.32



Measurement of fluid speed based on Bernoulli's principle.

(a) A manometer is connected to two tubes that are close together and small enough not to disturb the flow. Tube 1 is open at the end facing the flow. A dead spot having zero speed is created there. Tube 2 has an opening on the side, so the fluid has a speed *v* across the opening; thus, pressure there drops. The difference in pressure at the manometer is $\frac{1}{2}\rho v_2^2$, so *h* is proportional to $\frac{1}{2}\rho v_2^2$.

(b) This type of velocity measuring device is a Prandtl tube, also known as a pitot tube.

For an incompressible, frictionless fluid, the combination of pressure and the sum of kinetic and potential energy densities is constant not only over time, but also along a streamline:

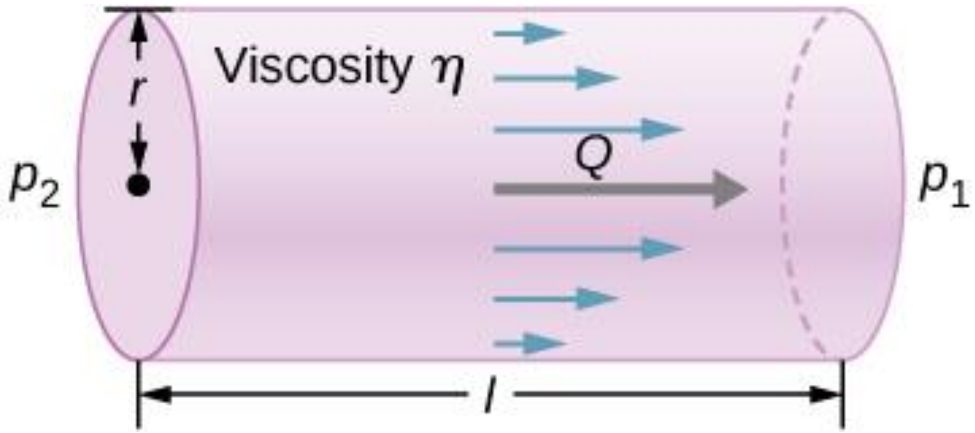
FIGURE 14.25

(a) Laminar flow can be thought of as layers of fluid moving in parallel, regular paths.

(b) In turbulent flow, regions of fluid move in irregular, colliding paths, resulting in mixing and swirling.

FIGURE 14.38

Poiseuille's law applies to laminar flow of an incompressible fluid of viscosity η through a tube of length *l* and radius *r*. The direction of flow is from greater to lower pressure. Flow rate *Q* is directly proportional to the pressure difference $p_2 - p_1$, and inversely proportional to the length *l* of the tube and viscosity η of the fluid. Flow rate increases with radius by a factor of r^4.



$$Q = \frac{p_2 - p_1}{R}$$

where p_1 and p_2 are the pressures at two points, such as at either end of a tube, and *R* is the resistance to flow. The resistance *R* includes everything, except pressure, that affects flow rate. For example, *R* is greater for a long tube than for a short one. The greater the viscosity of a fluid, the greater the value of *R*. Turbulence greatly increases *R*, whereas increasing the diameter of a tube decreases *R*.

The resistance *R* to laminar flow of an incompressible fluid with viscosity η through a horizontal tube of uniform radius *r* and length *l*, is given by

$$R = \frac{8\eta l}{\pi r^4}.$$

Poiseuille's law for resistance

Measuring Turbulence

An indicator called the **Reynolds number** N_R can reveal whether flow is laminar or turbulent. For flow in a tube of uniform diameter, the Reynolds number is defined as

$$N_R = 2\rho v r / \eta \quad \text{(flow in tube)}$$

where ρ is the fluid density, v its speed, η its viscosity, and r the tube radius. The Reynolds number is a dimensionless quantity. Experiments have revealed that N_R is related to the onset of turbulence. *For N_R below about 2000, flow is laminar. For N_R above about 3000, flow is turbulent*.

Example 14.8

Using Flow Rate: Air Conditioning Systems

An air conditioning system is being designed to supply air at a gauge pressure of 0.054 Pa at a temperature of $20\ ^\circ\text{C}$. The air is sent through an insulated, round conduit with a diameter of 18.00 cm. The conduit is 20-meters long and is open to a room at atmospheric pressure 101.30 kPa. The room has a length of 12 meters, a width of 6 meters, and a height of 3 meters. (a) What is the volume flow rate through the pipe, assuming laminar flow? (b) Estimate the length of time to completely replace the air in the room. (c) The builders decide to save money by using a conduit with a diameter of 9.00 cm. What is the new flow rate?

Example 14.9

Using Flow Rate: Turbulent Flow or Laminar Flow

In **Example 14.8**, we found the volume flow rate of an air conditioning system to be $Q = 3.84 \times 10^{-3}\ \text{m}^3/\text{s}$. This calculation assumed laminar flow. (a) Was this a good assumption? (b) At what velocity would the flow become turbulent?

Questions